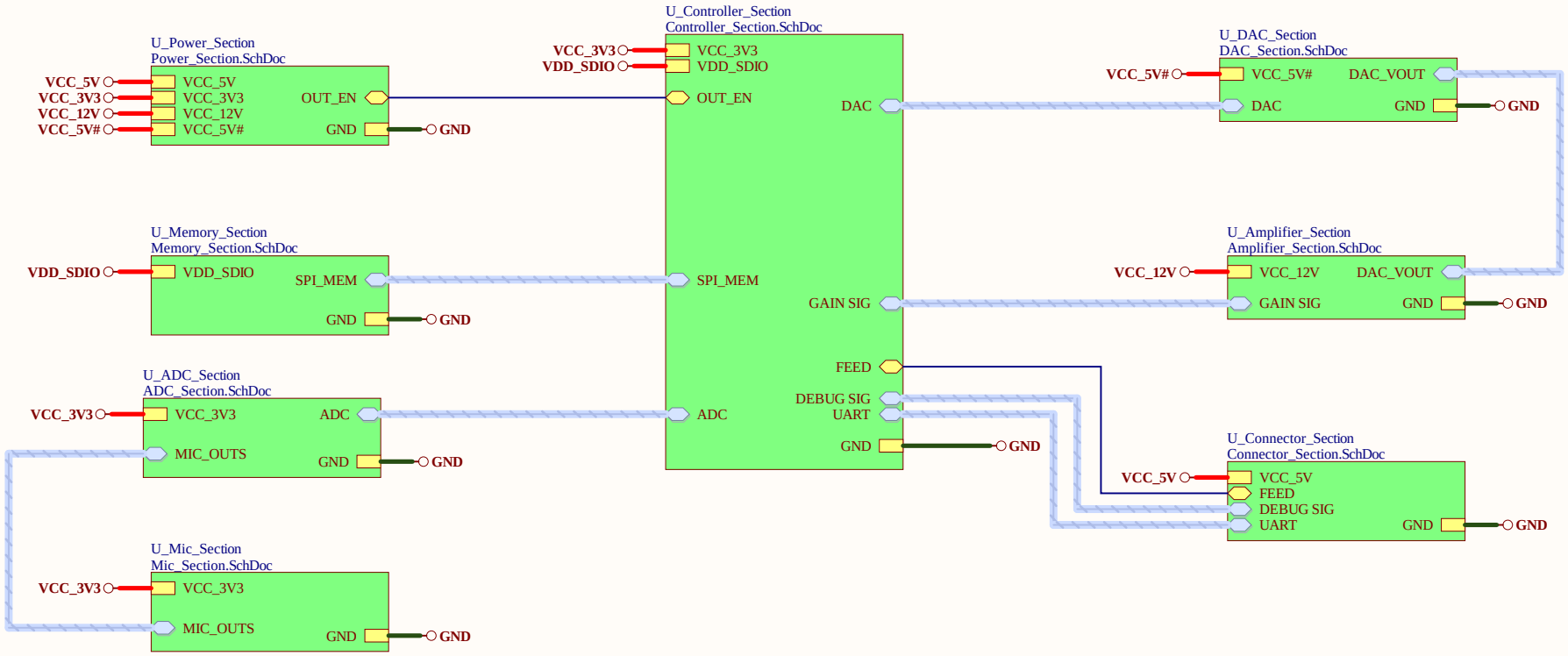
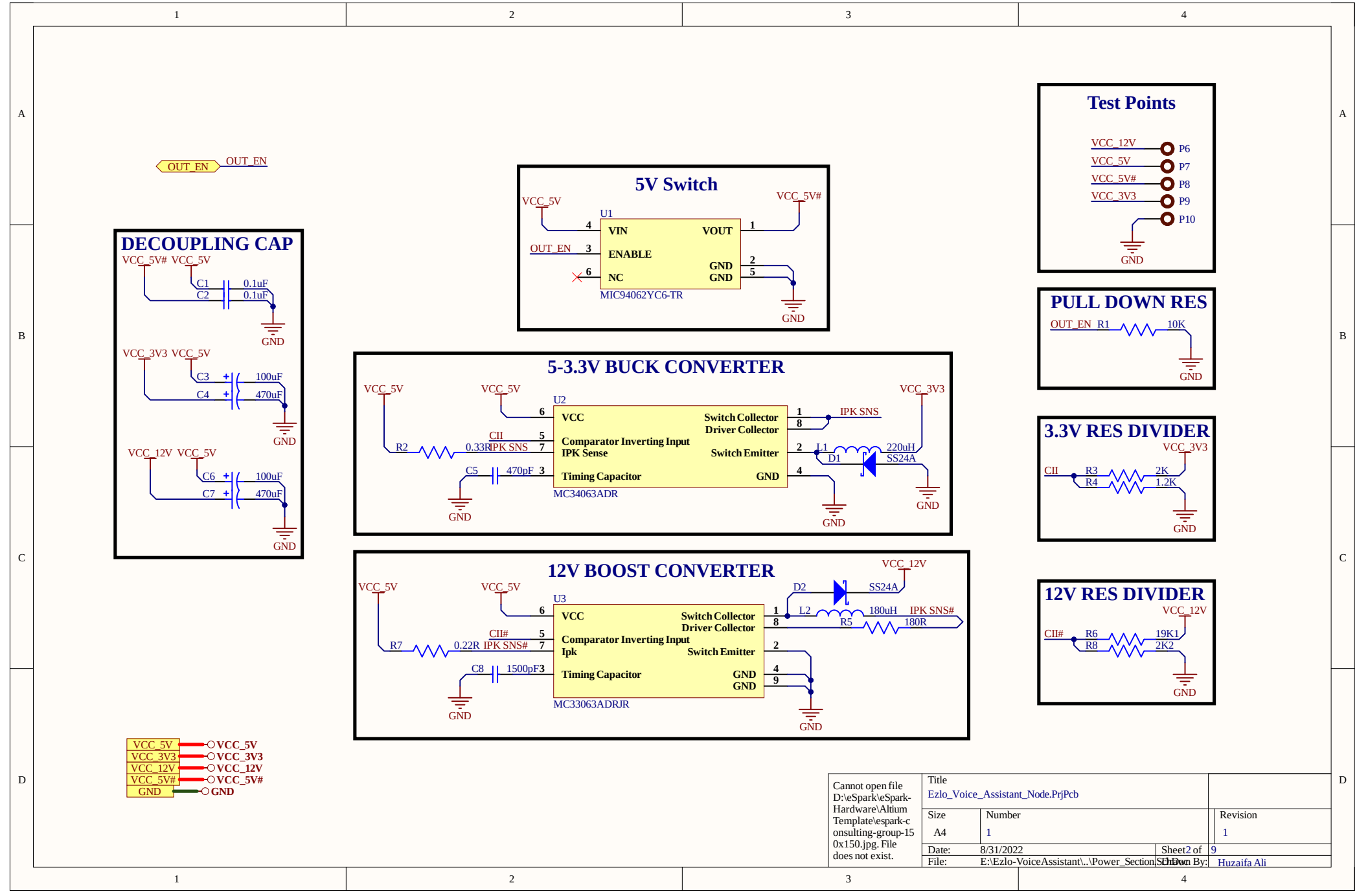
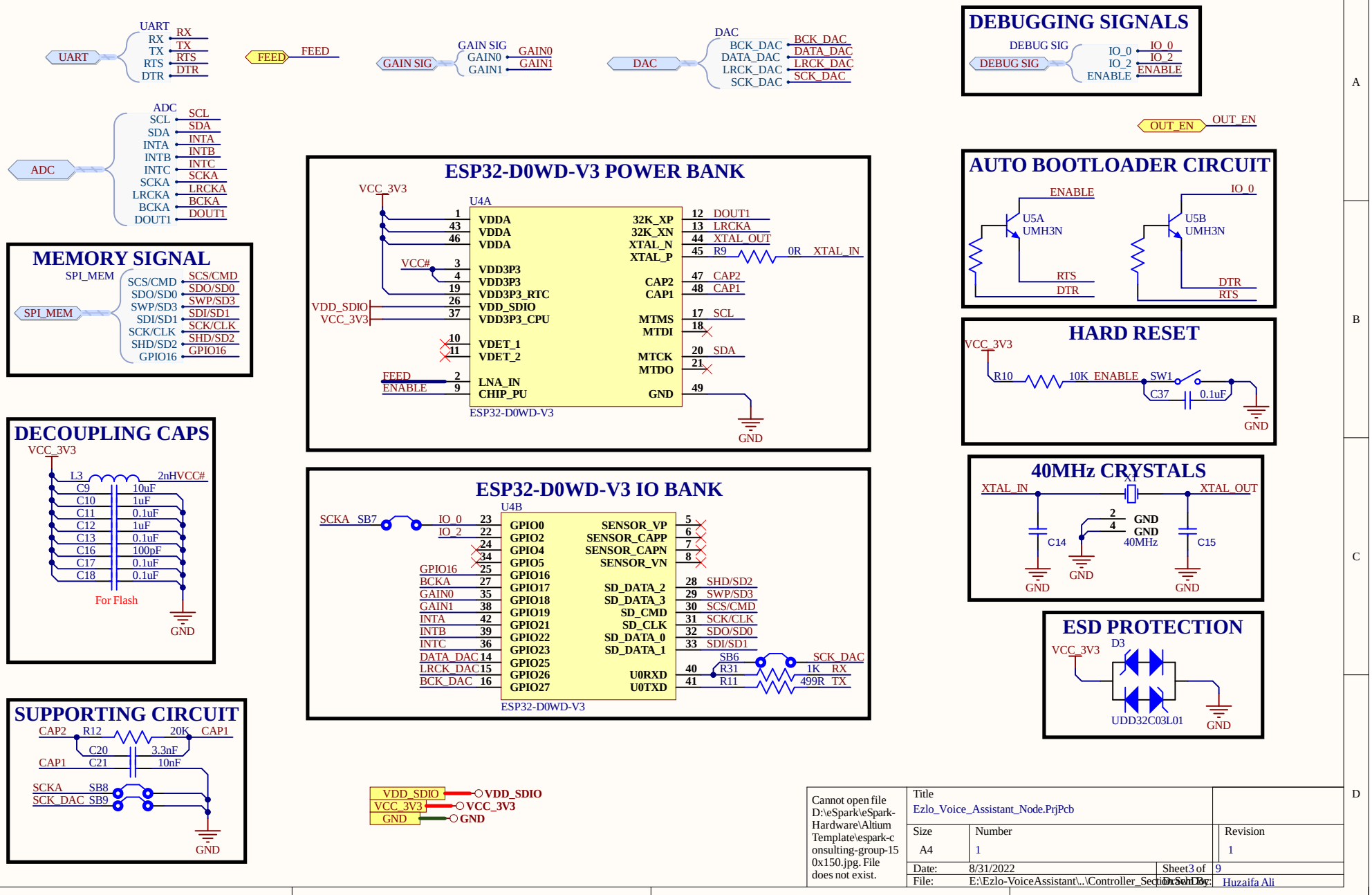


Cannot open file D:\eSpark\eSpark-Hardware\Altium Template\espark-consulting-group-150x150.jpg. File does not exist.

Title	Ezlo_Voice_Assistant_Node.PrjPcb		
Size	Number		Revision
A4	1		1
Date:	8/31/2022	Sheet 1 of	9
File:	E:\Ezlo-VoiceAssistant\..\Top_Sheet.SchDoc	Drawn By:	Huzaifa Ali

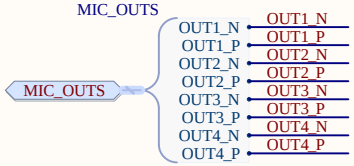


Cannot open file D:\eSpark\eSpark-Hardware\Altium Template\espark-consulting-group-150x150.jpg. File does not exist.	Title: Ezlo_Voice_Assistant_Node.PrjPcb		
	Size: A4	Number: 1	Revision: 1
	Date: 8/31/2022	Sheet 2 of 9	
	File: E:\Ezlo-VoiceAssistant\..\Power_Section.SchDoc	Drawn By: Huzaifa Ali	

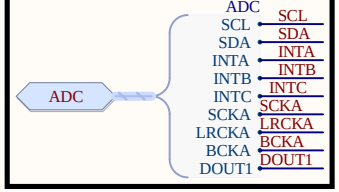


Cannot open file D:\eSpark\eSpark-Hardware\Altium Template\espark-consulting-group-150x150.jpg. File does not exist.	Title: Ezlo_Voice_Assistant_Node.PrjPcb	
	Size: A4 / Number: 1	Revision: 1
	Date: 8/31/2022	Sheet 3 of 9
	File: E:\Ezlo-VoiceAssistant\..\Controller_Section.SchDoc	Drawn By: Huzaifa Ali

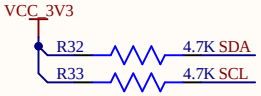
MIC OUT SIGNALS



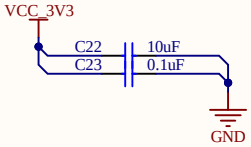
ADC SIGNALS



I2C Pull-up

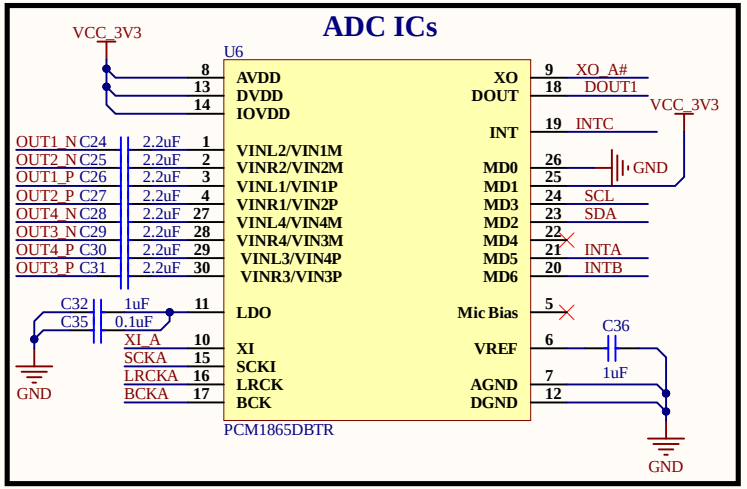


DECOUPLING CAPS

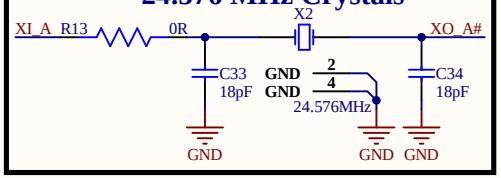


A minimum value of 2.2 µF is recommended in Figure 7 for codecs, which may have a very low input impedance at some PGA gain settings.
Datasheet page: 10

ADC ICs



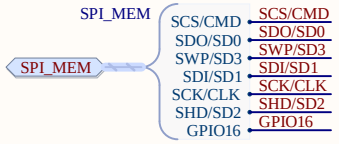
24.576 MHz Crystals



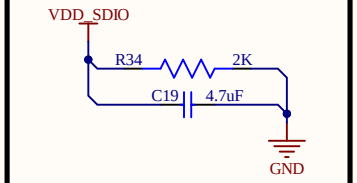


Cannot open file D:\eSpark\eSpark-Hardware\Altium Template\espark-consulting-group-150x150.jpg. File does not exist.	Title: Ezlo_Voice_Assistant_Node.PrjPcb	
	Size: A4 / Number: 1	Revision: 1
	Date: 8/31/2022	Sheet 4 of 9
	File: E:\Ezlo-VoiceAssistant\..\ADC_Section.SchDoc	Drawn By: Huzaifa Ali

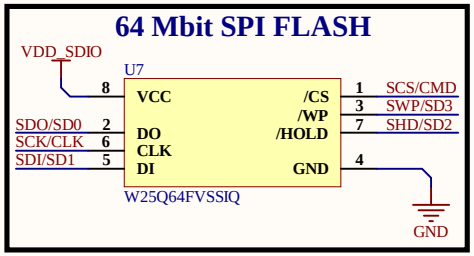
MEMORY SIGNAL



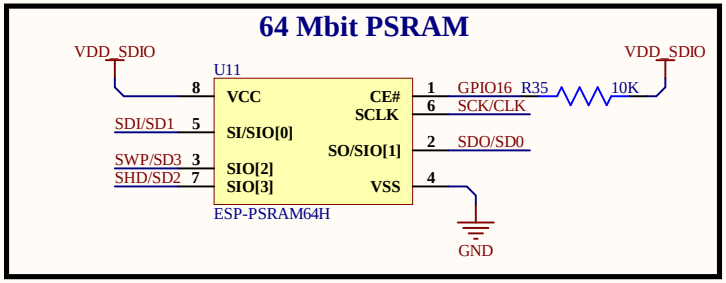
SUPPORTING CIRCUIT



64 Mbit SPI FLASH



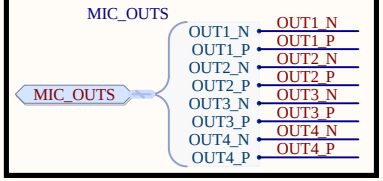
64 Mbit PSRAM



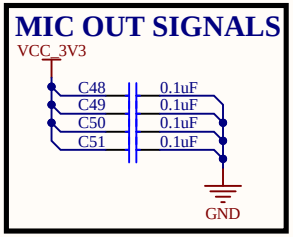


Cannot open file D:\eSpark\eSpark-Hardware\Altium Template\espark-consulting-group-150x150.jpg. File does not exist.	Title: Ezlo_Voice_Assistant_Node.PrjPcb		
	Size: A4	Number: 1	Revision: 1
	Date: 8/31/2022	Sheet 5 of 9	
	File: E:\Ezlo-VoiceAssistant\..\Memory_Section.SchDoc	Drawn By: Huzaifa Ali	

MIC OUT SIGNALS

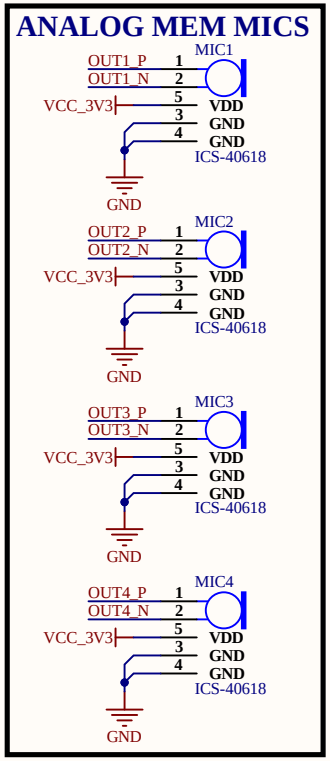


MIC OUT SIGNALS





ANALOG MEM MICS



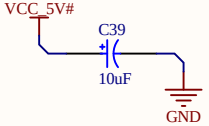
Cannot open file D:\eSpark\eSpark-Hardware\Altium Template\espark-consulting-group-150x150.jpg. File does not exist.	Title: Ezlo_Voice_Assistant_Node.PrjPcb		
	Size: A4	Number: 1	Revision: 1
	Date: 8/31/2022	Sheet 6 of 9	
	File: E:\Ezlo-VoiceAssistant\..\Mic_Section.SchDoc	Drawn By: Huzaifa Ali	



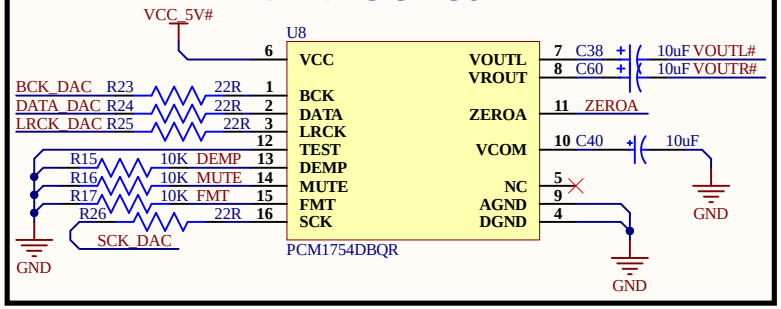


Input impedance of amplifier is 15K
when GAIN0-=0 and GAIN1=1
fC = 1/(2π × C × R)
Cutt of frequency of low pass filter is 10Hz
Range of sound signals= 20Hz-20KHz

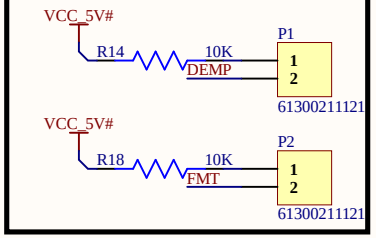
DECOUPLING CAP



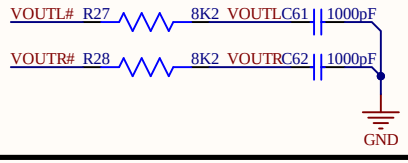
24 BIT DAC CIRCUITRY



FMT & DEMP CONTROL

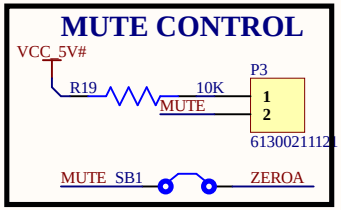


Supporting Circuitry



MUTE OFF - LOW
MUTE ON - HIGH
FMT LOW = 16– to 24–bit, I2S format
FMT HIGH =16–bit right–justified
DEMP HIGH = OFF
DEMP LOW = ON

MUTE CONTROL



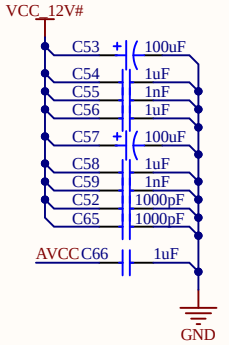


Cannot open file D:\eSpark\eSpark-Hardware\Altium Template\espark-consulting-group-150x150.jpg. File does not exist.	Title: Ezlo_Voice_Assistant_Node.PrjPcb	
	Size: A4 / Number: 1	Revision: 1
	Date: 8/31/2022 / Sheet 7 of 9	
	File: E:\Ezlo-VoiceAssistant\..\DAC_Section.SchDoc / Drawn By: Huzaifa Ali	

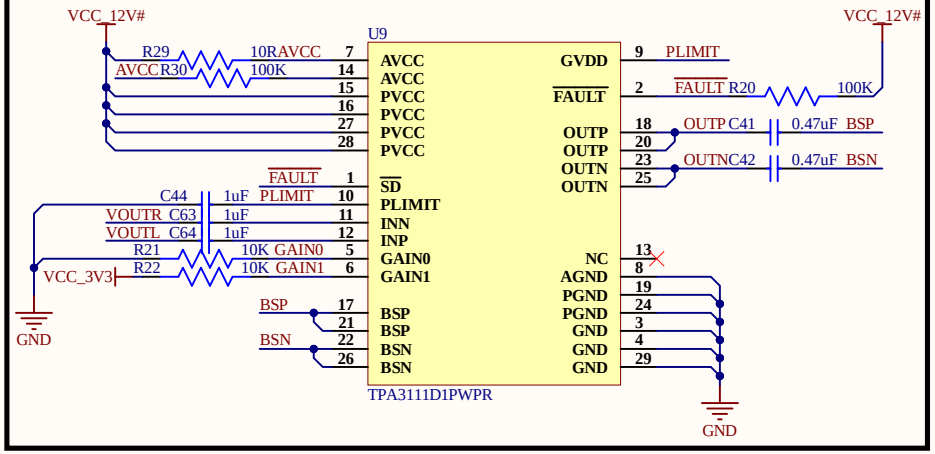
GAIN SIG
GAIN0
GAIN1

DAC_VOUT
VOUTL
VOUTR

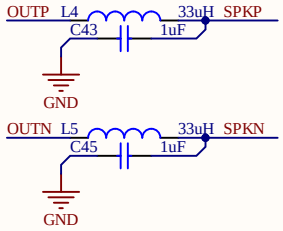
DECOUPLING CAP



10W 8OHMS AMPLIFIER IC



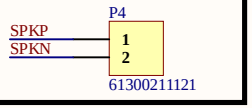
LPF FOR 8OHMs SPEAKER



Power filter



SPEAKER OUTPUT

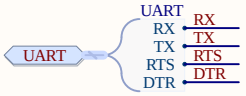


VCC_12V
GND

Cannot open file D:\eSpark\eSpark-Hardware\Altium Template\espark-consulting-group-150x150.jpg. File does not exist.	Title: Ezlo_Voice_Assistant_Node.PrjPcb		
	Size: A4	Number: 1	Revision: 1
	Date: 8/31/2022	Sheet 8 of 9	
	File: E:\Ezlo-VoiceAssistant\..\Amplifier_Section.SchDoc	Drawn By: Huzaifa Ali	



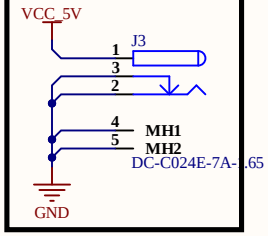
PROGRAM INPUTS



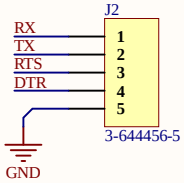
DEBUGGING SIGNALS



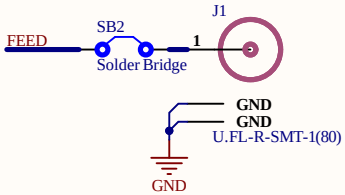
INPUT JACK



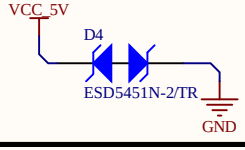
PROGRAMMING HEADER



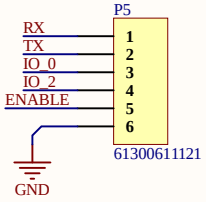
UFL CONECTOR EXTERNAL ANTENNA



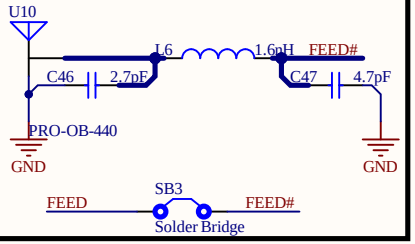
ESD PROTECTION



DEBUG CONNECTOR



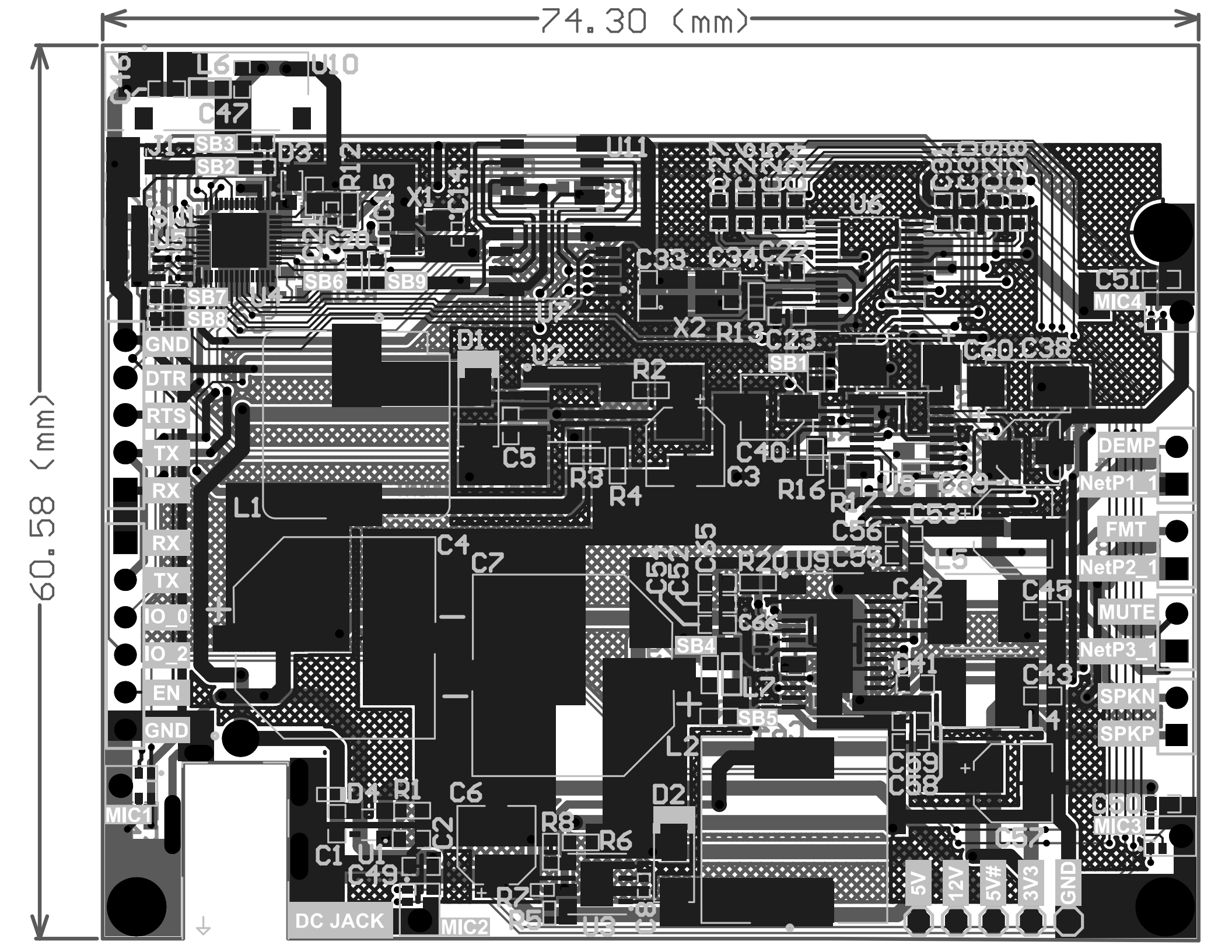
ON-BOARD PCB ANTENNA





Cannot open file D:\eSpark\eSpark-Hardware\Altium Template\espark-consulting-group-150x150.jpg. File does not exist.

Title	Ezlo_Voice_Assistant_Node.PrjPcb		
Size	Number		Revision
A4	1		1
Date:	8/31/2022	Sheet 9 of	9
File:	E:\Ezlo-VoiceAssistant\..\Connector_Section.SchDoc	Drawn By:	Huzaifa Ali


74.30 (mm)
60.58 (mm)
GND
DTR
RTS
TX
RX
RX
TX
IO_0
IO_2
EN
GND
DEMP
NetP1_1
FMT
NetP2_1
MUTE
NetP3_1
SPKN
SPKP
5V
12V
5V#
3V3
GND
DC JACK
MIC1
MIC2
MIC3
MIC4
SB1
SB2
SB3
SB4
SB5
SB6
SB7
SB8
SB9

Comment	Description	Designator	Footprint	LibRef	Quantity
0.1uF	CAP CER 0.1UF 50V 75V 0603, CAP CER 0.1UF 100V X7R 0603	C1, C2, C11, C13, C17, C18, C23, C35, C37, C45, C49, C50, C51	Cap0603	CC0603ZRY75V9BB104, CC0603KRX7R0BB104	13
100uF	25V 5mm 10mm 100uF 105kHz SMD,5x10mm Aluminum Electrolytic Cap- SMD ROHS	C3, C6, C53, C57	VKMB1003E101MV	VKMB1003E101MV	4
470uF	470uF ±20% 50V 12.5mm 13.5mm 10000hrs 105°C -40°C ~+105°C SMD,12.5x13.5mm Aluminum Electrolytic Capacitors - SMD RoHS	C4, C7	VT1H471M13500X030	VT1H471M13500X030	2
470pF	CAP CER 470pF 10% 4.7KV 50V 0603 SMD	C5	Cap0603	CC0603KRX7R9BB471	1
1500pF	CAP CER 1500PF 50V 10% 0402	C9	Cap 0402	CL05B152KB5NNNC	1
10uF	CAP CER 10UF 25V X5R 0603	C9, C22	Cap0603	C1206106MAMNYNC	2
1uF	CAP CER 1UF 25V X5R 0603, CAP CER 1UF 50V X7R 0603, CAP CER 1UF 16V X5R 0603	C10, C12, C32, C36, C50, C44, C45, C54, C55, C58, C63, C64, C66	Cap0603	CC0603KRX5R8BB105, UMK107AB7105KA-T, C1206106KAMNYNC, CC0603KRX5R7BB105	13
22PF	CAP CER 22PF 50V 0402NP0 0402	C14, C15	Cap0402	04025A220FAT2A	2
100pF	CAP CER 100PF 25V 0603	C16	Cap 0603	CC0603KRX7R8BB101	1
4.7uF	CAP CER 4.7UF 25V X5R 0603	C19	Cap0603	C1206475KAMNYNC	1
3.3nF	CAP CER 3300PF 50V 0402	C20	Cap0402	CGA3E2X7R1H332M050BA	1
10nF	CAP CER 10nF 50V X7R 0603	C21	Cap0603	CC0603KRX7R9BB103	1
2.2uF	CAP CER 2.2UF 25V X5R 0603	C24, C25, C26, C27, C28, C29, C30, C31	Cap0603	C1206225KAMNYNC	8
18pF	CAP CER 18PF 50V 0402NP0 0603	C33, C34	Cap0603	CC0603JRNP09BN180	2
10uF	CAP TANT 10UF 10% 16V 2312	C38, C60	Cap Case-C	T495C106K016AT	2
10uF	CAP ALUM 10UF 25V SMD	C39, C40	RVT1E100M0405	RVT1E100M0405	2
0.47uF	CAP CER 0.47UF 50V X5R 0603	C41, C42	Cap0603	C1206474KBMNNC	2
2.7pF	CAP SMD 2.7pF ±0.25pF 50V NPO 0603	C46	Cap 0603	CC0603CRNP09BWJ2R7	1
4.7pF	CAP SMD 4.7pF ±0.1pF 50V NPO 0603	C47	Cap 0603	CC0603BRNP09BWJ4R7	1
1000pF	CAP CER 1000PF 25V 0402NP0 0603	C52, C61, C62, C65	Cap0603	CC0603FRNP09BWJ102	4
1nF	CAP CER 1000PF 25V 0402NP0 0603	C56, C59	Cap0603	CC0603JRNP09BN102	2
S124A	60V 500mV@2A 2A SMA(DO-214AC) Schottky Barrier Diodes (SBD) ROHS	D1, D2	SMA	S124A	2
UDD3C03L01	15V 19A 3.3V 4V Bidirectional SOD-323/SC Protection Devices ROHS	D3	UDD10C03L01	UDD3C03L01	1
ESD5451N-2/TR	SOT1208-2L ESD Protection Devices ROHS	D4	DFN 1006-2L	ESD5451N-2/TR	1
LI-FL-R-SMT-1(80)	CONN LI-FL RCPT 17P 50 OHM SMD	J1	LI-FL-R-SMT-1(80)	LI-FL-R-SMT-1(80)	1
3-644456-5	35P MTA100 HDR ASSY (NARROW)	J2	HDR1X5	3-644456-5	1
DC-CD24E-7A-1.65	DC Power Receptacle 1.65mm 4.3mm 7A 24V Plugin AC/DC Power Connectors ROHS	J3	DC-CD24E-7A-1.65	DC-CD24E-7A-1.65	1
220uH	Inductor Power Shielded Wirewound 220uH 20% 1kHz Ferrite 1.3A 103mOhm DCR Blister T/R	L1	SLF12575T-221MSR3-PF	SLF12575T-221MSR3-PF	1
180uH	1.3A 180uH ±20% 350mΩ SMD 12x12x8mm Power Inductors ROHS	L2	FRHU1207-181M	FRHU1207-181M	1
2nH	2nH±0.3nH 0603 Inductors	L3	IND 0603	0DCL1608C2N0STDF	1
33uH	Fixed Ind 3320NUH 0.188 Ohm 1450M	L4, L5	SWPA5045S330MT	SWPA5045S330MT	2
1.6nH	700mA 1.6nH ±10% 40mΩ 0603 Inductors SMD ROHS	L6	IND 0603	PAW0603LC1N6KST	1
BLM18PG121SN1D	FERRITE BEAD 120 OHM 0603 1LN	L7	IND 0603	BLM18PG121SN1D	1
ICS-40618	MICROPHONE MEMS ANALOG OMNI	MIC1, MIC2, MIC3, MIC4	ICS-40618	ICS-40618	4
61300211121	CONN HEADER VERT 2POS 2.54MM	P1, P2, P3, P4	HDR1X2	61300211121	4
61300811121	CONN HEADER 8POS 2.54	P5	HDR1X8	61300811121	1
TEST POINT	Test Point ON	P6, P7, P8, P9, P10	TEST POINT	TEST POINT	5
10K	RES SMD 10K OHM 5% 1/10W 0603	R1, R10, R14, R15, R16, R17, R18, R19, R21, R22, R35	Res0603	AC0603JR-0710KL	11
0.33R	1Ω 1/10W Thick Film Resistors 75V ±600ppm/°C -55°C~+155°C 0.33Ω 0603 Chip Resistor - Surface Mount ROHS	R2	Res0603	RT1T0R330FTP	1
2K	RES SMD 2K OHM 1% 1/10W 0603	R3, R34	Res0603	RC0603FR-072KL	2
1.2K	RES 1.2K OHM 5% 1/10W 0603	R4	Res0603	MR06X1222TL	1
180R	±0.1% 1/16W Thin Film Resistor 25V ±25ppm/°C -55°C~+125°C 180Ω 0402 Chip Resistor - Surface Mount ROHS	R5	Res0402	AR02BTC1800	1
10R:1	RES 10.1K OHM 1% 1/10W 0603	R6	Res0603	RC0603FR-0710R1L	1
0.22R	800 mΩ 2Ω OHM 1% 1/8W 0402	R7	Res0402	PT0402FR-7W0R22L	1
2K2	RES SMD 2200Ω ±5.00% 0.1W ±200ppm/°C 0603	R8	Res0603	AC0603JR-072K2L	1
0R	RES SMD 0R, 1%, 0.1W, 0603	R9, R13	Res0603	RC0603FR-070RL	2
499R	RES SMD 499Ω ±1.00% 0.1W ±200ppm/°C 0603	R11	Res0603	RC0603FR-07499RL	1
20K	RES SMD 20K OHM 1% 1/10W 0603	R12	Res0603	RC0603FR-0720KL	1
100K	RES SMD 100K OHM 0.1% 1/10W 0603	R20	Res0603	RT0603BRD07100KL	1
22R	RES SMD 22 OHM 0.1% 1/10W 0603	R23, R24, R25, R26	Res0603	RT0603BRD07022RL	4
9K2	RES Thin Film SMD 9.2K OHM 0.1% 1/10W 0603	R27, R28	Res0603	RT0603BRD079K2L	2
10R	RES SMD 10 OHM 1% 1/4W 1206	R29	Res1206	RT1206FRD0710RL	1
100K	RES SMD 100K OHM 0.1% 1/8W 0805	R30	RES 0805	RT0805BRD07100KL	1
1K	RES SMD 1K OHM 0.5% 1/10W 0603	R31	Res0603	RT0603DRD071KL	1
4.7K	RES SMD 4.7K OHM 5% 1/10W 0603	R32, R33	Res0603	RC0603JR-074K7L	2
Solder Bridge	Solder Bridge ON	SB1, SB2, SB3, SB4, SB5, SB6, SB7, SB8, SB9	Solder Bridge	Solder Bridge	9
TS-1185EC-C-D-B	3mm 2mm Round Button 50mA Top3.65mm SPST 12V SMD Tactile Switches	SW1	TS-1185EC-C-D-B	TS-1185EC-C-D-B	1
MIC94062YC6-TR	IC PWR SWITCH P-CHAN 1:1 SCI5-6	U1	SC-70-6	MIC94062YC6-TR	1
MIC33063ADR	IC REG BUCK 85T ADJ 1.5A 8SOIC	U2	SOIC-8	MIC33063ADR	1
MIC33063ADRJR	IC REG BUCK 85T ADJ 1.5A 8SON	U3	MIC33063ADRJR	MIC33063ADRJR	1
ESP32-D0WD-V3	IC RF TXRX HALF-BLUETOOTH WIFI	U4	ESP32-D0WD-V3	ESP32-D0WD-V3	1
UMH3N	2 NPN pre-biased (Dual) 150mW 100mA 50V SC-70-6(SOT-363) Digital Transistors ROHS	U5	UMH3N	UMH3N	1
PCM1865DBTR	IC ADC/AUDIO 24BIT 192K 30TSSOP	U6	PCM1865DBTR	PCM1865DBTR	1
WZSQ64FV53IQ	IC SJSCH 64MBIT 80CHQUAD 8SOIC	U7	SOP-8_208mil	WZSQ64FV53IQ	1
PCM1754DBQR	IC DAC/AUDIO 24BIT 200K 16SSOP	U8	16-SSOP	PCM1754DBQR	1
TPA3111D2PWPR	Amplifier IC 1-Channel (Mono) Class D 28-HTSSOP	U9	28-HTSSOP	TPA3111D2PWPR	1
PRO-CIB-440	RF ANT 2.4GHZ IFAMPED MET SLD	U10	PRO-CIB-440	PRO-CIB-440	1
ISP-PSRAM64H	IC PSRAM 64MBIT SPI 133MHZ 8SOIC	U11	SOIC-8	4077	1
40MHz	40MHz SMD Crystal Resonator 20pF ±10ppm ±30ppm -40°C~+85°C SMD3225-4P Crystals ROHS	Y1	X322540MPB4SI	X322540MPB4SI	1
24.576MHz	24.576MHz SMD Crystal Resonator 10pF 60Ω ±10ppm -40°C~+85°C SMD3225-4P Crystals ROHS	X2	0322457621060	0322457621060	1